Beyond Short Names: Cross-Model Emoji Lexical Neighborhoods with No Evidence for a Simple Late-Layer Mediator

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Abstract

Emoji reach text language models as Unicode token sequences, not rendered pictures. We ask whether a raw atomic emoji indexes only its official short name or a wider lexical neighborhood, and whether that relation has a simple late-layer mediator. In a preregistered observational study, we cross 55 emoji and 55 non-emoji Unicode symbols with raw, canonical-name, lexically disjoint CLDR-keyword, and literal codepoint views. Across nine frozen model/tokenizer cells, the emoji alias-minus-codepoint retrieval effect is positive in both unrestricted and exact four-view token-count-matched analyses. All nine cells exceed a 0.02 registered effect-size floor; the alias-correspondence randomization primary passes with four positive core cells and partial-conjunction $p = 0.0027$. Canonical names are also retrieved for symbols, but symbol alias effects are smaller and heterogeneous in this registered secondary comparison. Passing the observational gate admits a separate, more stringent causal test. We learn rank-8 subspaces on 128 source emoji, then intervene at six late attention and MLP outputs for 192 emoji held out from T19-L in Llama 3.2 3B. All engineering controls reproduce the baseline exactly, yet no primary site passes the joint necessity–sufficiency gate. Together, these differently gated results narrow the explanatory space: the text-only emoji-to-CLDR lexical relation is repeatable, but the preregistered sparse late-component account is not supported.

1 Introduction

Emoji look like compact visual objects to readers, but a text language model receives them as encoded Unicode scalars and tokenizer-dependent sequences. That distinction matters for interpretation. A representation probe may recover an official label because the label co-occurs with the character in standards, documentation, or training data. Such a result need not imply rendered-glyph perception, a human-like concept, or a localized semantic circuit.

We therefore ask a narrower two-part question. First, does a raw atomic emoji retrieve a lexically disjoint CLDR keyword better than its literal `U+...` spelling after exact token-count matching? Second, if this observational relation is reliable, can a frozen low-rank late-component intervention identify a representation that is both necessary and sufficient for it? We do not claim that models "understand emoji." The contribution is an auditable separation between a broad lexical readout and one simple mechanistic account.

The study is organized around three safeguards and their corresponding results:

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- A four-view, three-bank protocol separates raw Unicode surface, official English CLDR name, a word-disjoint CLDR keyword, and literal codepoint spelling. The causal source and evaluation banks are held out from T19-L, although their surfaces may have appeared in studies before T19.
- The preregistered observational primary passes: 4/4 core cells and 9/9 total cells are positive in unrestricted and exact four-view token-count matched analyses. The cross-model partial-conjunction value is 0.0027.
- A conditionally admitted causal experiment passes all zero-hook and restoration controls but finds no primary block-24 attention or MLP output that is jointly necessary and sufficient. A secondary block-27 MLP patch is sufficient-only and is not a complete mediator.

The observational result extends beyond exact-short-name retrieval under the frozen endpoint, while the causal result does not support one preregistered rank-8 late-component mechanism. Neither result identifies which of the remaining distributed, redundant, nonlinear, or item-private accounts is correct.

2 Related work

Emoji as lexical objects. Prior work learns emoji embeddings from descriptions [Eisner et al., 2016] and builds sense inventories from lexical and usage resources [Wijeratne et al., 2016]. We instead treat CLDR names and keywords [Unicode Consortium, 2026] as fixed, source-auditable endpoints for probing frozen hidden states. The goal is not to produce a better emoji embedding, but to test which textual relation is recoverable from a raw Unicode input and which alternative explanations remain.

Probing and tokenizer controls. Probe accuracy can reflect properties of the control task rather than the target representation [Hewitt and Liang, 2019]; tokenization itself can materially change language-model comparisons [Rust et al., 2021]. We avoid a learned diagnostic classifier, use item-wise retrieval, and require both unrestricted and exact four-view token-count-matched tests. This is a token-count control, not token-identity matching or tokenizer independence. Our readout is also more local than whole-representation similarity analyses such as CKA [Kornblith et al., 2019]: it asks whether the correct item is the nearest cross-view partner.

Causal intervention. Causal mediation and activation interventions can distinguish readable correlates from components that affect model computation [Vig et al., 2020, Meng et al., 2022, Turner et al., 2023]. We adopt that distinction but require a frozen site to pass both random-adjusted necessity and sufficiency on a disjoint evaluation bank. The negative result is retained rather than followed by an adaptive search over layers, ranks, or endpoints.

3 Prospective design

3.1 Claim boundary, views, and banks

For auditability, we retain two internal study labels. **T19-L** denotes the observational lexical-retrieval arm ("L" for lexical), and **T19-C** denotes the conditionally admitted component-causal arm. They are labels for the two arms defined below; no knowledge of the preceding project sequence is required. Appendix A gives the short decision lineage that led to T19-L without treating earlier exploration as part of its confirmatory evidence.

Throughout, *glyph* means the raw Unicode scalar string supplied to a text model. No raster image is rendered. Atomic emoji are single-codepoint, fully-qualified Emoji 17 entries introduced no later than Emoji 13.1 [Unicode Consortium, 2025]. For each item, the canonical view is the exact English CLDR 48.2 `tts` value; the alias is one exact English CLDR `default` keyword. After lowercasing and extracting ASCII `[a-z0-9]+` words, canonical and alias word sets must be nonempty and disjoint. The fourth view is the uppercase literal codepoint spelling. When several keywords qualify, aliases are assigned by one deterministic maximum bipartite matching with globally unique labels; a frozen SHA-256 edge order breaks ties before any T19 model forward.

Raw scalar†	CLDR `tts`	Keyword	Literal view
U+1F923	rolling on the floor laughing	lmao	`U+1F923`
U+1F928	face with raised eyebrow	skeptic	`U+1F928`
U+00B4	acute accent	tone	`U+00B4`

Table 1: Concrete frozen views for two emoji and one non-emoji symbol. †The model receives the scalar itself; the table names it by codepoint because the paper typeface need not render every glyph. The literal view is the displayed ASCII string.

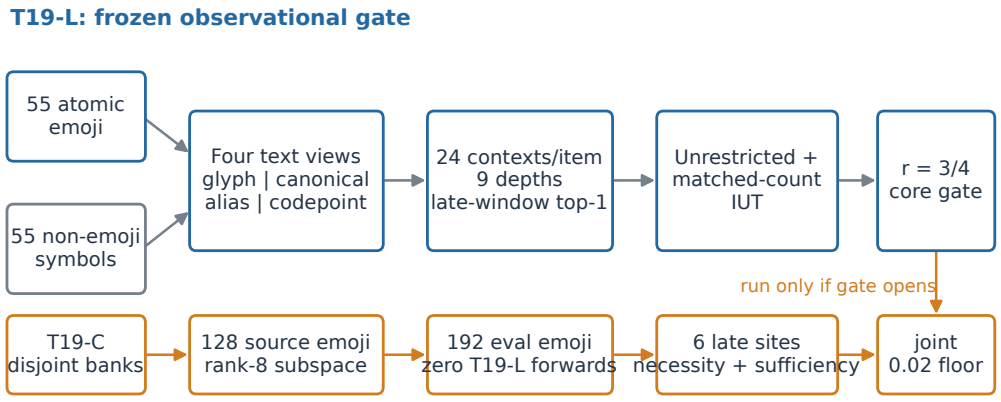


Figure 1: Prospective T19 design. T19-L crosses two glyph kinds with four text views. Only the emoji endpoint enters the registered cross-model primary. T19-C runs only after that gate opens and uses disjoint source and evaluation emoji. No causal-bank item receives a T19-L forward.

The observational bank contains 55 emoji and 55 non-emoji Unicode symbols. Symbols are single scalars in a Unicode general category beginning with `S`, have `Emoji=false`, have valid disjoint CLDR aliases, and are absent from every strict-UTF-8, no-NUL repository blob at the frozen boundary. This is repository-relative freshness, not pretraining novelty. Exact emoji surfaces may have appeared in earlier experiments; the endpoint, alias assignment, and bank split are prospective, not the emoji surfaces.

Two further banks contain 128 causal-source and 192 causal-evaluation emoji. All 430 codepoints and 430 aliases are unique across banks. Figure 1 summarizes the separation and the one-way admission gate.

3.2 Observational readout

For each item and view, the frozen harness appends one of two suffix carriers (`" a"` or `" |"`) at distances 1, 2, 4, and 8 and records the hidden state at the selected suffix token at nine normalized depths. Batch size is one; no chat template or added special token is used. The primary late window is fixed at depths 0.75, 0.875, and 1.0, yielding 24 contexts per item.

Within each view and context, hidden states are centered across items and each row is L2-normalized. Let $z_i^{v,c}$ denote the resulting vector for item i, view v, and context c. Cross-view top-1 retrieval is

$$R_c(g \to v) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\left[i = \arg\max_j \left\langle z_i^{g,c}, z_j^{v,c} \right\rangle \right]. \tag{1}$$

The per-model primary estimate is

$$\Delta_m = \frac{1}{24} \sum_{c} \left[R_c(g \to \textsc{alias}) - R_c(g \to \textsc{codepoint}) \right] \tag{2}$$

on the 55 observational emoji. This is a nearest-neighbor readout from hidden states, not a next-token generation score.

3.3 Token-count intersection and cross-model test

Every model must pass two components. The unrestricted component uses all 55 emoji and permutes alias correspondence within glyph kind. The matched component retains non-singleton strata sharing the exact four-view token-count signature and permutes aliases only within stratum. Retrieval in that component uses the retained pool ($n = 31$–41 emoji across models), so its chance level is $1/n$; unrestricted and matched magnitudes are not treated as directly commensurable. Canonical and codepoint correspondences stay fixed. Thus the one-sided randomization test asks whether the frozen alias mapping beats mappings allowed by its null; because the codepoint term is fixed, it is not a paired significance test of alias–codepoint equality. The registered primary combines that test with the observed comparative requirement $\Delta_m \geq 0.02$ in both components. At $n = 55$, the latter requires at least two net top-1 successes; in the smaller matched pools it requires at least one. Each component uses 19,999 draws, and the model intersection-union p-value is the maximum of the two component values.

The sole cross-model primary uses four frozen core cells: Llama 3.2 3B base, Qwen2.5 7B base, OLMo 2 1B base, and Gemma 2 2B IT. It tests the partial conjunction that at least $r = 3$ of $m = 4$ cells pass [Benjamini and Heller, 2008]. With ordered model IUT values $p_{(1)} \leq \cdots \leq p_{(4)}$, the registered value is $p_{\mathrm{PC}} = \min(1, 2p_{(3)})$. The other five cells and all symbol or canonical-name results are secondary and cannot open the causal gate.

We report percentile intervals from 20,000 item bootstrap replicates as descriptive uncertainty. They are not substituted for the permutation-based primary decision.

3.4 Conditionally admitted causal test

After the observational gate was committed, T19-C ran in the frozen Llama 3.2 3B base cell. At each of six sites (attention and MLP component outputs at blocks 21, 24, and 27), we learned a rank-8 basis from the left singular directions of the centered, row-normalized glyph-alias cross-covariance on 128 source emoji. A deterministic rank-8 control basis was drawn from the same source span and made orthogonal to the learned basis. Blocks 24 attention and MLP were primary; blocks 21 and 27 were secondary.

On 192 different evaluation emoji, necessity removes learned coordinates from the glyph component output. Sufficiency adds the same-item learned glyph-minus-codepoint coordinates to the codepoint output. Analogous operations in the random basis define adjusted effects:

$$N_s = E_s(\text{random removal}) - E_s(\text{learned removal}), \tag{3}$$

$$S_s = E_s(\text{learned patch}) - E_s(\text{random patch}), \tag{4}$$

where E_s is the final-hidden alias-minus-codepoint top-1 endpoint. A site must have both effects and both lower 95% item-bootstrap bounds above 0.02, pass both one-sided sign-flip tests (99,999 draws), pass zero and remove-restore engineering controls, and survive Holm correction across the two primary sites [Holm, 1979]. No site, rank, depth, carrier, subset, or endpoint may be replaced after seeing the result.

4 Results

4.1 A lexical relation beyond the official short name

The registered cross-model primary passed. All four core cells were positive; the third-smallest core IUT value was 0.00135, yielding $p_{\mathrm{PC}} = 2(0.00135) = 0.0027$. All nine frozen cells were positive and exceeded the registered 0.02 effect-size floor in both unrestricted and matched analyses (Figure 2; exact values in Appendix C). Model IUT values ranged from 0.00005 to 0.00185. An independent implementation reconstructed all 36 glyph-kind by component randomization tests and the same gate decision.

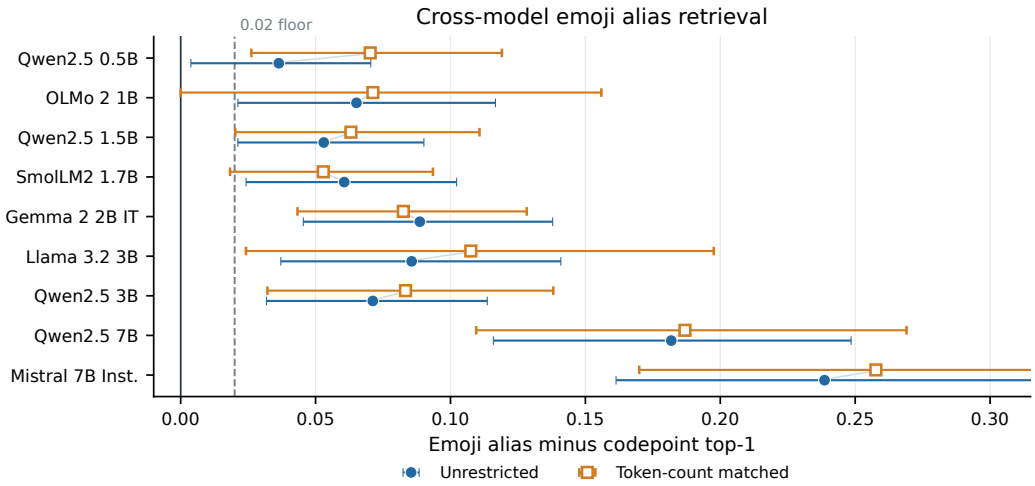


Figure 2: Emoji alias-minus-codepoint top-1 effects in the registered roster order. Bars are descriptive 95% item-bootstrap intervals (20,000 replicates); the randomization tests determine the primary decision. Every point estimate exceeds the registered 0.02 floor in unrestricted and exact token-count-matched analysis.

Across models, mean emoji alias effects were $+0.0979$ unrestricted and $+0.1084$ matched. Mean canonical-name effects were larger, $+0.1440$ and $+0.1718$. Thus, retrieval extends beyond the exact official name, although the official name remains the stronger key on average. The alternative label is an exact CLDR keyword, not a human-rated synonym or paraphrase; training-data co-occurrence could still fully explain the relation.

Depth was secondary. The nine recorded fractions were $0, 0.125, \ldots, 1.0$; only 0.75, 0.875, and 1.0 formed the primary window. Across models and contexts, the unrestricted contrast was $+0.0409$ at depth 0, then $+0.0864$, $+0.1152$, and $+0.0922$ at the three primary depths, whose mean is the $+0.0979$ reported above. Individual cells had different onset and peak depths, so these data do not support one universal trajectory.

4.2 A named-symbol control limits the emoji-specific claim

Non-emoji symbols showed mean alias effects of $+0.0236$ unrestricted and $+0.0153$ matched, compared with $+0.0979$ and $+0.1084$ for emoji. The emoji-minus-symbol point estimate was positive in all nine cells in both components. Yet symbol aliases were heterogeneous: Qwen2.5 3B and 7B were negative in both components. Canonical-name retrieval exceeded alias retrieval for symbols in 9/9 cells in both components, with mean symbol canonical effects of $+0.0750$ and $+0.0821$ (Figure 3).

This comparison cannot establish an emoji-specific faculty. The panels are not matched on training frequency, lexical specificity, Unicode category, visual familiarity, or token identity. Its role is to prevent the stronger but false inference that only emoji acquire readable names.

4.3 No frozen late-component site passes both causal gates

Before intervention, the final-hidden glyph endpoint on the 192-item causal evaluation bank was $E_s = 0.07292$ at every site (alias top-1 0.07292; codepoint top-1 0). All six causal sites then passed their engineering checks: zero hooks and remove-restore conditions reproduced the baseline final hidden state with exactly zero RMS difference. Nevertheless, neither primary block-24 site passed. Block-24 attention had adjusted necessity 0.0000 and sufficiency -0.00521; block-24 MLP had $+0.00521$ and 0.0000. Their site IUT values were 1.0 and 0.74912, respectively. None of the six sites passed both components (Figure 4; exact values in Appendix D).

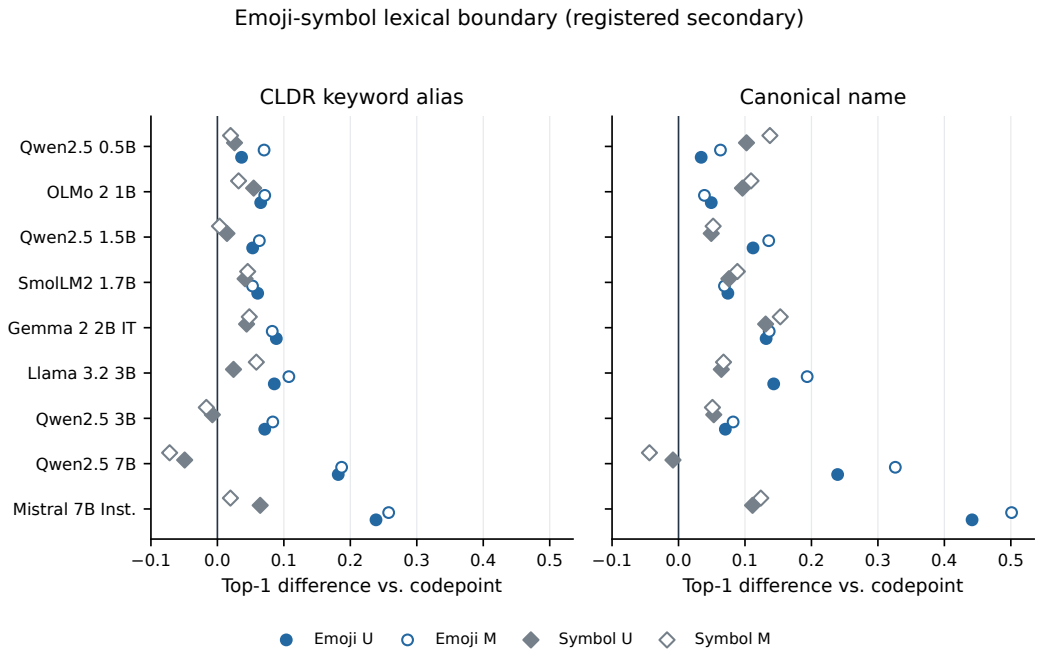


Figure 3: Registered secondary comparison. Each row shows unrestricted (U) and token-count-matched (M) effects for the same frozen model. Official-name retrieval is not exclusive to emoji, while disjoint-keyword effects are stronger and more consistent for the frozen emoji panel. Glyph category, frequency, label quality, and token identity are not exchangeable, so this is a boundary result rather than an emoji-only semantic claim.

The secondary block-27 MLP patch produced adjusted sufficiency $+0.07812$ (95% interval $[+0.03646, +0.11979]$, one-sided uncorrected $p = 0.00009$ across 12 component tests), but necessity was -0.01562 (interval $[-0.04167, +0.00521]$, $p = 0.96895$). The site IUT was therefore 0.96895. Moreover, the learned patch raised alias top-1 from 0.0052 to 0.0208 while lowering codepoint self-retrieval from 1.0000 to 0.9375. It is an incomplete perturbation effect, not clean alias insertion or a necessary mechanism.

5 Discussion

The observational and causal results answer different questions. T19-L shows that, under a fixed text-only readout, the same item is recoverable across raw emoji and a lexically disjoint CLDR keyword more reliably than across raw emoji and a literal codepoint string. Exact token-count matching makes a pure length account insufficient. The symbol panel shows that official-name readout is a broader Unicode phenomenon, while keyword breadth is graded rather than categorically emoji-only.

T19-C does not support a tempting mechanistic shortcut: a shared rank-8 late attention or MLP component, learned from one bank and tested on another, does not pass joint necessity and sufficiency at the frozen sites. This failure does not adjudicate the observational measurement; the two arms test different claims. It instead shows that aggregate linear readability does not entail one sparse causal mediator. Distributed, redundant, nonlinear, and item-private accounts remain possible, but the present data do not distinguish among them.

The block-27 MLP asymmetry sharpens that distinction. A patch can move an endpoint without identifying the normal route that produces it. Requiring necessity, sufficiency, random-subspace adjustment, restoration, and a disjoint evaluation bank prevents that isolated patch success from being upgraded to a circuit claim.

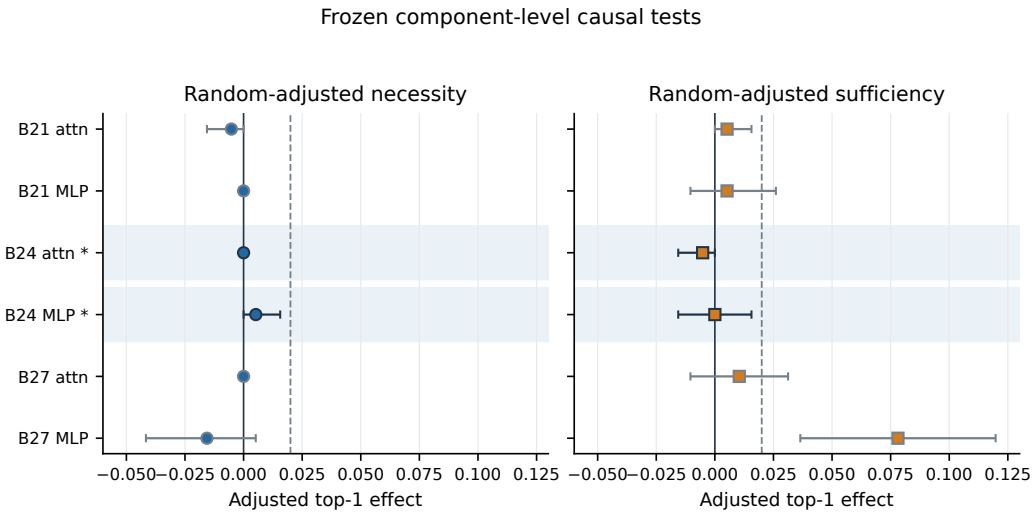


Figure 4: Random-adjusted causal effects with descriptive 95% item-bootstrap intervals over 192 emoji held out from T19-L. Shaded rows and asterisks mark the two primary block-24 sites. Solid vertical lines show zero; dashed lines show the registered 0.02 effect-size floor. No site has both necessity and sufficiency above the floor with positive lower bounds.

6 Limitations and broader impact

Measurement. CLDR keywords are standardized annotations, not human judgments of synonymy, natural naming, specificity, or familiarity. Retrieval is nearest-neighbor geometry at selected suffix tokens, not generation or task behavior. Inputs are Unicode text; no rendered glyph is perceived. Exact token-count matching does not match token identities or equalize tokenizer construction.

Generalization. The nine model/tokenizer cells are a fixed, convenience-constrained roster spanning several families and sizes [Yang et al., 2024, Groeneveld et al., 2024, Dubey et al., 2024, Gemma Team et al., 2024, Jiang et al., 2023, Hugging FaceTB, 2024]. Model family, parameter count, training data, tokenizer, and instruction tuning are confounded, so the study supports neither a scale law nor a model-population estimate. Exact emoji surfaces are not pretraining-novel. The symbol panel is repository-text-fresh but not a pretraining holdout and is not exchangeable with the emoji panel.

Mechanism. The negative causal result covers one Llama 3.2 3B checkpoint, rank 8, one carrier and distance, two primary and four secondary sites, and one final-hidden endpoint. Its exact engineering gates verify hook fidelity, not power: no frozen positive intervention control calibrates the dynamic range at the two primary sites. It does not show that no causal representation exists. A future study should preregister such a positive control, distributed multi-site removal, nonlinear or sparse readouts, and a new source/evaluation split rather than selecting a method on the present evaluation bank.

Reproducibility and impact. The anonymous compact archive binds configurations, selections, summaries, tests, validators, license notices, and the rendered paper, but omits model-dependent raw arrays retained in execution scratch. Exact acquisition duration and peak memory were not receipt-bound and are not reconstructed after the fact. Better Unicode auditing could reveal hidden representation failures in multilingual or social-media systems. Conversely, treating CLDR correspondence as a person's meaning, sentiment, culture, or identity could erase contextual variation and enable unjustified profiling. The measurements should not be used for such inference. No personal data, human subjects, or user messages are used.

7 Conclusion

Across nine frozen model/tokenizer cells, raw atomic emoji retrieve word-disjoint CLDR keyword aliases better than literal codepoints after exact four-view token-count matching. A named-symbol control bounds the specificity of that observation, and a separately banked causal experiment finds no primary rank-8 late component that is jointly necessary and sufficient. The two results narrow different parts of the explanation: broad lexical readability is repeatable, but it should not be mistaken for visual understanding or evidence for a simple localized mechanism.

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A Exploration path and decision lineage

T19-L is the nineteenth internal study label in a longer audit trail, not a concept that readers are expected to know in advance. The concise lineage below includes only results that changed the next design decision. These studies motivated T19-L, but they are not pooled with its preregistered primary, and their exploratory or failed-confirmation status is unchanged.

Initial screening and token controls (baseline and Milestone 2). A colored-shape steering screen produced 25 positive results among 36 layer–seed–dose cells, but an exact tokenization-matched follow-up supported one tested layer contrast and not the other. We therefore made tokenization part of the estimand, retained negative cells, and stopped treating a pooled steering score as one stable semantic effect.

Lexical exploration and failed reserves (T9–T12). A held-out Mistral cell first supported name-over-codepoint retrieval. A nine-cell follow-up retained statistical signal but failed its registered cross-model effect-size rule. Broader glyph-system exploration then suggested Unicode-composition hypotheses, but neither survived testing on separately reserved items. We kept both failures, dropped the generic composition claim, and returned to a narrower lexical question.

Readable responses and unsuccessful causal tests (T15–T16). An emoji–word response pattern replicated prospectively, and frozen forced-choice associations were detectable. Yet a subspace intervention and a multi-layer transplantation each failed their stronger causal gates. This separated readout from implementation and motivated the present joint necessity–sufficiency criterion.

Depth mapping and fresh confirmation (T17–T18). Across 292 previously exposed atomic emoji, name-minus-codepoint retrieval was positive in all nine model cells at middle-to-late depths. Peaks were model-specific, and final item retrieval (9/9) did not require consistent whole-representation CKA alignment (4/9 model cells). We therefore froze a common late window, replaced the population, and made no shared-transition or universal-manifold claim. On 192 repository-fresh composite emoji, the official-name contrast then passed in 4/4 core cells and 8/9 cells overall after exact token-count matching. This justified asking whether the relation extends beyond the official short name, while adding a named non-emoji control and reserving new banks for causality.

T19-L and T19-C (current confirmatory and conditional arms). T19-L replaces the sole label endpoint with a word-disjoint CLDR keyword and adds literal-codepoint and named-symbol boundaries. T19-C tests the frozen rank-8 component account on disjoint source and evaluation banks. The present paper reports the resulting pair of differently gated results: broad lexical retrieval passes, while the simple late-component mediator does not.

This lineage explains why the final design contains four text views, exact token-count strata, a non-emoji symbol panel, three disjoint item banks, and a necessity–sufficiency intersection. It also explains what is absent: no favorable prior layer, glyph system, or causal site was promoted into the T19-L primary. Only that frozen primary supports the paper's confirmatory lexical claim.

B Frozen model roster

Table 2 reports the exact model artifacts and revisions used by the frozen observational roster. The Llama role uses the pinned BF16 mirror artifact specified by the protocol. Despite its `mlx-community` identifier, the frozen run loaded its safetensors through the Transformers/CUDA backend; we make no upstream bit-identity claim. Model cards and upstream license terms are part of the submission-archive license audit. The cards declare Apache-2.0 for the Qwen2.5 0.5B, 1.5B, and 7B cells, OLMo, SmolLM2, and Mistral; Qwen2.5 3B uses the non-commercial Qwen Research License, Gemma uses the Gemma Terms of Use, and the Llama mirror uses the Llama 3.2 Community License. No checkpoint or tokenizer file is redistributed.

Role	Artifact identifier	Revision
Qwen2.5 0.5B base	`Qwen/Qwen2.5-0.5B`	`060db6499f32`
OLMo 2 1B base	`allenai/OLMo-2-0425-1B`	`a1847dff3500`
Qwen2.5 1.5B base	`Qwen/Qwen2.5-1.5B`	`8faed761d45a`
SmolLM2 1.7B base	`HuggingFaceTB/SmolLM2-1.7B`	`effd688a1292`
Gemma 2 2B IT	`google/gemma-2-2b-it`	`299a8560bedf`
Llama 3.2 3B base	`mlx-community/Llama-3.2-3B-bf16`	`60a99aaf4316`
Qwen2.5 3B base	`Qwen/Qwen2.5-3B`	`3aab1f1954e9`
Qwen2.5 7B base	`Qwen/Qwen2.5-7B`	`d14972939875`
Mistral 7B Inst. v0.3	`mistralai/Mistral-7B-Instruct-v0.3`	`c170c708c41d`

Table 2: Frozen artifact identifiers and the first 12 hexadecimal characters of their complete pinned revisions. Full revisions are stored in the submission manifest.

C Exact observational results

Frozen cell	U effect	M effect	Model IUT p	Pass
Qwen2.5 0.5B	0.0364	0.0702	0.00055	yes
OLMo 2 1B	0.0652	0.0712	0.00135	yes
Qwen2.5 1.5B	0.0530	0.0631	0.00185	yes
SmolLM2 1.7B	0.0606	0.0528	0.00005	yes
Gemma 2 2B IT	0.0886	0.0825	0.00005	yes
Llama 3.2 3B	0.0856	0.1075	0.00135	yes
Qwen2.5 3B	0.0712	0.0833	0.00005	yes
Qwen2.5 7B	0.1818	0.1869	0.00005	yes
Mistral 7B Inst.	0.2386	0.2577	0.00005	yes

Table 3: Exact reporting values for the registered emoji alias-minus-codepoint endpoint. U is unrestricted; M is exact four-view token-count matched. The model IUT is the maximum component randomization p-value.

Matched emoji counts ranged from 31 to 41. The matched component retained at least 63 observational items total in every cell, with at least 31 emoji and 32 symbols; the unrestricted component used all 55 emoji and 55 symbols. The acquisition comprised 7,920 forwards, 285,120 readout vectors, and 45 sealed shards.

D Exact causal results

Block	Site	Primary	N_s	p_N	S_s	p_S	Site IUT
21	attention	no	-0.00521	1.00000	0.00521	0.49983	1.00000
21	MLP	no	0.00000	1.00000	0.00521	0.49760	1.00000
24	attention	yes	0.00000	1.00000	-0.00521	1.00000	1.00000
24	MLP	yes	0.00521	0.49989	0.00000	0.74912	0.74912
27	attention	no	0.00000	1.00000	0.01042	0.31511	1.00000
27	MLP	no	-0.01562	0.96895	0.07812	0.00009	0.96895

Table 4: Random-adjusted necessity (N_s) and sufficiency (S_s). All six engineering gates passed. Holm correction applies across the two primary sites and did not change either decision; no site passed the component-level intersection.

The causal run used 256 source forwards, 2,880 evaluation forwards, and 8,064 intervention rows. Each of 12 component tests used 99,999 sign flips and 20,000 item-bootstrap replicates. An independent validator rehashed source and evaluation arrays, reconstructed all 12 tests and intervals, verified learned and random subspace relations, and reproduced six engineering passes and zero positive sites.

E Execution, licensing, and artifact boundary

Both observational and causal acquisition were frozen to a Transformers/CUDA BF16 backend and an NVIDIA GeForce RTX 5090. Model artifacts, tokenizer replay receipts, selection files, analysis summaries, randomization seeds, bootstrap seeds, and validators are checksum-bound. Figure inputs in the paper are only the compact validated model-cell and causal-site JSON summaries; a generated manifest hashes both inputs and every plotted PDF/PNG output.

The anonymous evidence archive is a deterministic whitelist-only zip. Its manifest records every payload path, byte count, and SHA-256; a validator rejects unsafe paths, author or local-path tokens, model-weight formats, and raw array formats. The archive includes the paper, compact results, selected banks, code, validation receipts, and paired license/compute/ethics reports. It omits model-dependent activation arrays, learned causal subspaces, the full intervention ledger, model weights, and tokenizer files. Those raw scientific objects remain bound by run receipts in execution scratch. The package thus supports compact-summary and figure audit, not self-contained raw replay.

The observational acquisition used 7,920 batch-one forwards and produced 285,120 readout vectors. Causal acquisition used 256 source forwards, 2,880 evaluation forwards, and 8,064 intervention rows. No training or generation was performed. Exact wall-clock time, peak memory, energy use, and total exploratory compute were not preserved in the T19 compact receipts; the compute checklist therefore remains negative rather than substituting older run timings.

Selected Unicode 17 and CLDR 48.2 rows are used under Unicode License v3 and travel with its copyright and permission notice. New project code is Apache-2.0. Restricted checkpoint assets are referenced by pinned revision and upstream terms rather than redistributed. In particular, the Qwen2.5 3B cell is used only for non-commercial research under the Qwen Research License.

We reviewed the study against the NeurIPS Code of Ethics. It uses no human participants, personal data, private messages, or scraped user content. LLM coding and research assistants supported protocol drafting, code generation, manuscript editing, and one author-side fresh-eyes review; they did not provide labels, choose outcomes after T19 execution, or serve as the scientific evaluator. The authors retained execution authority and approved the frozen endpoints and final claims; numerical evidence and validators were checksum-bound.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: Sections 1, 5, and 6 separate the positive observational result, the negative registered causal result, and their bounded interpretation.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 6 discusses measurement, tokenization, model-roster, causal-scope, archive, and social-interpretation limits.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper reports empirical tests and does not introduce a theoretical result.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [No]

Justification: Sections 3 and A–E disclose the protocol, decision lineage, frozen revisions, code, and compact evidence, but the anonymous package omits model-dependent activation arrays and the full intervention ledger needed for exact independent raw replay.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [No]

   Justification: N/A does not apply because the experiments require code. A checksum-bound anonymous compact artifact is supplied, but it omits the model-dependent raw arrays required for exact independent replay. Full code and eligible data are planned for public release after review.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 3 specifies all evaluation banks, views, contexts, readouts, tests, gates, interventions, and fixed sites; no model training is performed.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Sections 3.3–3.4 define the randomization, partial-conjunction, sign-flip, Holm, and bootstrap procedures; Figures 2 and 4 label bootstrap intervals as descriptive.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [No]

   Justification: Appendix E identifies the RTX 5090 and BF16 CUDA path and reports forward counts, but exact wall-clock and peak-memory receipts are not in the compact archive.

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10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: Section 6 discusses auditing benefits and the risk of converting standardized lexical correspondence into unjustified cultural, sentiment, or identity inference.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The work releases no new pretrained model, scraped dataset, or high-risk capability; it studies public Unicode annotations and existing model checkpoints.

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Answer: [Yes]

Justification: Appendices B and E and the archive license register identify every model and data term. Model/tokenizer files are not redistributed; the Qwen 3B cell is bounded to non-commercial research; and the Unicode License v3 notice accompanies the derived rows.

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Justification: Appendix E conservatively discloses assistant support for protocol drafting, code generation, editing, and fresh-eyes review, while separating it from labels, frozen decisions, execution authority, numerical evidence, and scientific evaluation.